

Sharp Moment Reconstruction at Surreal Scales

Collision geometry, optimal precision, and Hahn-supported inversion

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Abstract

We study recovery of a finite weighted configuration from its first $2n$ power moments over real and complex Hahn fields, and hence over the surreal and surcomplex numbers. For distinct finite nodes a_i and nonzero weights w_i , put $d_i = \sum_{j \neq i} v(a_i - a_j)$ and $\delta_i = \max_{j \neq i} v(a_i - a_j)$. We prove that

$$\Theta = \max_i (v(w_i) + 2d_i + \delta_i)$$

is the exact uniform valuation threshold for guaranteeing reconstruction inside all strict nearest-neighbour balls. Errors of valuation at least $\kappa > \Theta$ admit a unique regular n -node realization, with node error at least $\kappa - v(w_i) - 2d_i$ and weight error at least $\kappa - 2d_i - \delta_i$. Perturbing only the last moment attains every node bound simultaneously; at valuation Θ it defeats the full strict-ball guarantee for every starting configuration. We also give an arbitrary-rank precision-lattice theorem, quantify cancellation in the inverse Hermite matrix, and derive a directed-path certificate for nonuniform moment precisions. The arguments use finite algebra and strong Hahn summability, not an Archimedean contraction argument. Classical Prony reconstruction, Hermite interpolation, and differential precision tracking are explicitly separated from the proposed refinements. Priority is not certified; the article supplies proofs rather than claiming a solution to a named published conjecture.

In this collection the report is built from a single manuscript and filed under Prony’s name, because “moment” has other meanings in neighbouring reports. Section 1.4 records its provenance, fixes the meaning of “moment” used here, and compares Θ with the coefficient threshold $T(P) = \max_i (d_i + \delta_i)$ of the collection’s polynomial-algebra report.

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1 The problem, the contribution, and its scope

A finite atomic configuration is specified by distinct nodes $a_1, \dots, a_n$ and nonzero weights $w_1, \dots, w_n$. Its moments are

$$m_k = \sum_{i=1}^n w_i a_i^k, \quad 0 \leq k < 2n. \quad (1)$$

Section 1.4 fixes this meaning of “moment” for the whole report and compares it with other uses of the word in the collection. For ordinary complex data this is the classical Prony problem. When two nodes are extremely close, inversion magnifies measurement errors. Over surreal scalars, closeness can occur at several incomparable Archimedean orders: one cluster may be separated at scale ω^{-1} , another at $\omega^{-\omega}$, and nested clusters may combine both. A single real condition number cannot retain that ordered-group information.

The question addressed here is precise: *how much moment precision forces the reconstructed nodes to retain their individual labels and their entire finite separation geometry?* We do not merely seek continuity or a first-order condition number. We seek a sufficient threshold, exact node losses above it, and a counterexample at the threshold itself.

Throughout, “regular n -node realization” means exactly n distinct nodes with nonzero weights. Uniqueness is in this fixed-order model; $2n$ moments do not determine arbitrary configurations with more than n nodes.

1.1 Statement of the main result

Let $K = \mathbb{k}((t^\Gamma))$, where $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$ and Γ is a nontrivial ordered abelian group. The support of a Hahn series is well-ordered, and $v(t^\gamma) = \gamma$. Write $\mathcal{O} = \{x : v(x) \geq 0\}$ and $\mathfrak{m} = \{x : v(x) > 0\}$, with $v(0) = +\infty$. Assume $a_i \in \mathcal{O}$, and define

$$\begin{aligned} d_{ij} &= v(a_i - a_j), & d_i &= \sum_{j \neq i} d_{ij}, & \delta_i &= \max_{j \neq i} d_{ij}, \\ \varpi_i &= v(w_i), & E_i &= \varpi_i + 2d_i, & \Theta &= \max_i (E_i + \delta_i). \end{aligned} \quad (2)$$

For $n = 1$ the empty sum and the empty maximum in this notation are both assigned the value zero. The convention fixes the residue ball in that case. The letters d_{ij} , d_i and δ_i have exactly the meaning they have in the collection’s polynomial-algebra report; the weight valuations are written ϖ_i , because α_i denotes roots there (Section 1.4).

Theorem 1.1 (Optimal uniform reconstruction). *Suppose $\hat{m}_k = m_k + \epsilon_k$, with $v(\epsilon_k) \geq \kappa > \Theta$ for $0 \leq k < 2n$. There is a unique regular n -node realization of the perturbed moments, up to permutation. It has a unique labelling satisfying*

$$v(\hat{a}_i - a_i) > \delta_i. \quad (3)$$

With this labelling,

$$v(\hat{a}_i - a_i) \geq \kappa - E_i, \quad (4)$$

$$v(\hat{w}_i - w_i) \geq \kappa - 2d_i - \delta_i > \varpi_i. \quad (5)$$

Consequently the leading terms of the weights and of every nonzero pairwise node difference are preserved.

If only m_{2n-1} is perturbed, by a nonzero e of valuation $\kappa > \Theta$, equality holds in (4) for every i . If instead $v(e) = \Theta$, no regular n -node realization of those moments can satisfy (3) for all i .

The final assertion means that the open precision condition is genuinely optimal for *this* labelling guarantee. It does not assert that every boundary perturbation causes a multiple root, or that reconstruction is impossible below the threshold. Some perturbations have much more cancellation than the worst case.

1.2 Relation to the literature

Prony reconstruction by a Hankel system, polynomial roots, and a Vandermonde system is classical. Batenkov and Yomdin study local accuracy of confluent Prony systems [1]. Katz, Diab, and Batenkov obtain sharp clustered complex reconstruction estimates and analyze correlations between the successive errors [2]. In particular, their first-order node formula already has a denominator consisting of the weight times squared pairwise separations. Thus the occurrence of $w_i P'(a_i)^2$ below is not presented as a previously unknown conditioning mechanism. For a single cluster of n unit-weight nodes with all separation valuations equal to s , our formulas give the familiar node loss $(2n - 2)s$ and threshold $(2n - 1)s$.

The refinement proposed here is the *exact ordered-group statement* for arbitrary finite nested separation patterns, weights at arbitrary valuations, and an optimality witness valid at every configuration. The threshold is pointwise rather than a cluster-size asymptotic. It applies without embedding the value group into $\mathbb{R}$ and without restricting nodes to an ordinary unit circle. The proof isolates the correlations by changing to the cofactor basis before making any estimates.

There is also a classical geometric antecedent to the sharpness direction: Akinshin–Goldman–Yomdin study Prony varieties obtained by fixing initial moments, including the curve on which the first $2n - 1$ moments stay fixed [3]. Thus varying only the last measured moment is not claimed as a newly discovered direction. The proposed refinement is its exact valuation consequence for every arbitrary-rank configuration, together with the uniform converse.

Exact propagation of precision lattices by a differential is an established non-Archimedean method, notably in Caruso–Roe–Vaccon [4]. Our Section 7 is a finite-polynomial, arbitrary-rank version proved by Hahn support evaluation. Neither its general philosophy nor Hermite interpolation is claimed as new. Its use here provides exact nonlinear uncertainty sets and exposes a cancellation invariant not recorded by the separation tree alone. The nonuniform certificate of Section 8 combines the cofactor system with directed-path bounds; it is a sufficient certificate, not an exact characterization.

1.3 Relation to the supplied repository

The comparison uses the repository snapshot

4cf691c7d951e037739d32d9f5c387dcce724f3c.

The documentation catalogue, formalization-ledger excerpts, and the polynomial-algebra report’s detailed README were inspected [7]. That report already contains finite Hermite jets, Hahn factor lifting, root-cluster geometry, and a *coefficient-perturbation* threshold of the form $\max_i(d_i + \delta_i)$. (In that report’s current labels these are `polynomial:thm:crt`, the section `polynomial:sec:hensel`, the cluster tree of `polynomial:subsec:tree`, and `polynomial:thm:stability` [8].) Those are

antecedents, not contributions of this article. The present input is a vector of *moments*, whose structured inversion changes the threshold to $\max_i(\varpi_i + 2d_i + \delta_i)$ and requires control of the recovered weights. Remark 6.2 states the relation between the two thresholds precisely.

At the pinned snapshot, repository searches for “Prony” and “Hankel” returned no indexed matches, and the separate `docs/new` tree listed eighteen ZIP archives; their binary contents and all historical revisions were not exhaustively read. Accordingly, “not found in the inspected repository material” is the scope of the nonduplication check. Both observations are provenance, not current descriptions. The first remained true of the collection’s reports until this batch was placed, but the collection now also contains the Hankel and Prony-type material compared in Section 1.4. The archives that were then in `docs/new`, and the later ones including the archive that delivered this manuscript, have since been placed into reports and removed from that directory. The literature comparison is likewise not a proof of priority. The strongest candidate contribution is Theorem 1.1 together with its universal last-moment sharpness mechanism; the other statements are useful supporting refinements.

1.4 This report in the collection

Provenance. This report is built from one manuscript: the article “Sharp Moment Reconstruction at Surreal Scales”, dated 22 September 2026, with its verification program and records. It was manuscript 13 of the eighteen placed in the collection’s eighteenth batch, delivered as the archive `surcomplex_moment_reconstruction`, and its repository comparison is pinned to commit `4cf691c` (the snapshot above). It is not a merge: no other manuscript proves its main results, and the nearest material in the collection, including a classical Prony-type lemma, is compared below. On placement the prefix `prony`: was added to every label, and the four symbols in the table below were renamed. This subsection, Section 6.2, a note after Proposition 3.1, pointers to other reports and two ledger rows in Appendix A were added. Statements about the repository that were true only at the pin are now marked as such (Sections 1.3 and 10 and Appendix B). The only choices were editorial: the manuscript’s title is kept, the directory is named after Prony, and the notation follows the polynomial-algebra report. Otherwise the statements, proofs, limitations and priority caveats are those of the manuscript, and the program `code/verify.py` and the files in `data/` are unchanged from the delivery.

Here	Manuscript	Meaning	In polynomial-algebra
d_{ij}	s_{ij}	$v(a_i - a_j)$	d_{ij} , the same
δ_i	r_i	$\max_{j \neq i} d_{ij}$	δ_i , the same
ϖ_i	α_i	$v(w_i)$	not used; α_i are roots there
χ_i	h_i	$\ell'_i(a_i) = \sum_{j \neq i} (a_i - a_j)^{-1}$	not used

The sum d_i has the same meaning in both reports. The letter χ_i also keeps the Hermite coefficient apart from $h = t^s$ in Section 9. The program keeps the manuscript’s letters `alpha`, `r` and `hi`. Throughout, Θ is the threshold (2), an element of Γ ; it has nothing to do with the theta series of the collection’s hahn-tate-uniformization report. $T(P)$ always denotes the polynomial-algebra threshold; the letter T inside the proofs of Proposition 7.2 and Theorem 8.1 is local.

What “moment” means here. In this report a moment is always one of the $2n$ power moments (1) of a finite weighted configuration, and the input of reconstruction is that finite vector. The

weights are arbitrary nonzero elements of K ; no measure and no positivity is involved, and Corollary 5.4 only records that positivity is preserved when it is present. Two neighbouring reports use the word for other objects: `surreal/hahn-valued-measures-and-probability` [10] for power moments of Hahn-valued measures and moment functionals, including infinite moment sequences and their representation problem, and `surcomplex/hahn-herglotz-positivity` [11] for Fourier (trigonometric) moments on the circle, tested by Toeplitz matrices. Power sums of roots, that is, unit-weight moments with repeated roots allowed, are the “contour moments” of `d:thm:moments` in `surcomplex/analysis` [12] and the Newton sums of the Hankel section discussed below. This is why the directory carries Prony’s name rather than the word “moment”.

The coefficient threshold. Theorem 1.1 is the moment-input analogue of `polynomial:thm:stability` [8], whose input is a coefficient perturbation of a monic polynomial with distinct finite roots and whose threshold is $T(P) = \max_i (d_i + \delta_i)$. Remark 6.2 shows that $\Theta = \max_i ((\varpi_i + d_i) + (d_i + \delta_i))$, where $\varpi_i + d_i = v(w_i P'(a_i))$ is the cost of passing from moments to the annihilator, and that Lemma 4.1 is a nodewise form of the matching step of that theorem.

Hankel matrices. With unit weights, the matrix H of Proposition 3.1 is the Hankel matrix of the Newton power sums of P , that is, the matrix $VV^\top$ of the trace form of $K[X]/(P)$, and $\det H = \text{Disc } P$. This classical identity is the only thing this report shares with the section on Hankel square-class profiles in `surcomplex/spectral-theory` [9], which comes from another manuscript of the same batch and states the identity as its `spec:hank:prop:hermite`. That section reads square classes, modulo 2Γ , of the leading principal minors of such matrices over nondivisible value groups. This report estimates valuation thresholds for perturbed weighted moments and uses neither square classes nor leading principal minors. The two share no theorem, neither uses the other, and they are cross-referenced, not merged.

The measures report. `surreal/hahn-valued-measures-and-probability` [10] contains, from another manuscript of the same batch, a Prony-type lemma on finite signed exponential sums (a real sequence is a finite sum $\sum_j c_j x_j^n$ with distinct x_j exactly when a monic polynomial with distinct roots annihilates it under the shift) and a finite Gaussian quadrature theorem over real closed fields (a suitably positive linear functional agrees up to degree $2N - 1$ with N positive weights at N distinct nodes). Those results concern existence and exact representation of moment data, under positivity. This report starts from an n -node configuration and studies the stability of its reconstruction, with exact thresholds, and involves no measure. The results are cross-referenced, not merged.

2 Hahn fields, support, and surreal interpretation

2.1 Conventions and two different notions of smallness

An element of K is a formal sum

$$x = \sum_{\gamma \in S} x_\gamma t^\gamma, \quad x_\gamma \in \mathbb{k},$$

with $S \subset \Gamma$ well-ordered. The valuation of a nonzero series is its least exponent. Coefficient-zero extraction is a homomorphism $\text{st} : \mathcal{O} \rightarrow \mathbb{k}$ with kernel $\mathfrak{m}$. Every $x \in \mathcal{O}$ has a unique expression

$x = x_0 + x_+$ with $x_0 \in \mathbb{k}$ and $x_+ \in \mathfrak{m}$. For vectors, $v_{\min}(x) = \min_j v(x_j)$.

A family is *strongly summable* when the union of its supports is well-ordered and every exponent belongs to only finitely many supports. The Hahn sum is then defined coefficientwise. This is not a claim that finite partial sums converge in the full surreal fine topology. Nor does $\eta > 0$ imply that $N\eta$ eventually exceeds every element of Γ . For example, in $\mathbb{Q}^2$ with lexicographic order, $N(0, 1) < (1, 0)$ for every integer N . All infinite constructions used below are justified by support, not by such a false cofinality assertion.

Lemma 2.1 (Positive-support evaluation). *If $z_1, \dots, z_q \in \mathfrak{m}$ and $F \in \mathbb{k}[[Z_1, \dots, Z_q]]$, its monomial expansion can be evaluated strongly at $(z_1, \dots, z_q)$. Evaluation preserves addition, multiplication, and formal substitution when the substituted series have zero constant term. The resulting support is contained in the additive monoid generated by $\bigcup_j \text{supp}(z_j)$.*

Proof. Let S be the finite union of the positive well-ordered supports, omitting zero inputs. The word form of Higman's well-quasi-ordering theorem [6] implies the usual positive-support lemma: S^* , the monoid of finite sums from S , is well-ordered, and a fixed exponent is the sum of only finitely many finite words in S . Here is the deduction. Order words by subsequence embedding with each letter allowed to increase. An embedding never decreases the sum, and increases it strictly unless the words are identical. An infinite decreasing sequence of sums, or an infinite collection of distinct words with the same sum, would therefore be a bad sequence for the word order. Higman's theorem excludes both. The empty word supplies the sole sum of length zero.

For each input monomial, expand its product of Hahn series into words. There are only finitely many choices of input labels for a fixed word. The preceding finiteness is therefore exactly the local finiteness needed to evaluate F . It also justifies the double-family regroupings in products and zero-constant substitutions. No bound on the ordinary magnitudes of the coefficients of F is needed, because their valuations are zero. $\square$

Lemma 2.2 (Residue-simple lifting). *If $f \in \mathcal{O}[Y]$, $c \in \mathbb{k}$, and $\bar{f}(c) = 0$, $\bar{f}'(c) \neq 0$, there is a unique $y \in c + \mathfrak{m}$ with $f(y) = 0$. The same conclusion holds for a finite polynomial system whose residue Jacobian at $c \in \mathbb{k}^N$ is invertible.*

Proof. Write every coefficient as its residue plus an infinitesimal, and temporarily replace the finitely many infinitesimal coefficients by independent formal parameters Z . In the ring $\mathbb{k}[[Z]]$, solve for $y = c + u(Z)$ with $u(0) = 0$ by total degree. At each positive degree the only contribution containing the as-yet-unknown homogeneous part of u is the invertible constant Jacobian times that part. All other terms have already been determined. This gives a unique formal solution. Evaluate it by Lemma 2.1.

For uniqueness in $c + \mathfrak{m}^N$, subtract two equations. Polynomial divided differences give a matrix multiplying their difference; its residue is the invertible residue Jacobian. Its determinant is a unit of $\mathcal{O}$, so the difference is zero. This also proves uniqueness without an iterative-limit argument. $\square$

2.2 Finite families in the full surreal field

We use the standard Conway normal-form interpretation

$$\sum_{\gamma} c_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} c_{\gamma} \omega^{-\gamma}.$$

For any set-sized ordered subgroup $\Gamma \subset \mathbf{No}$, real Hahn series with exponents in Γ identify with the corresponding normal forms; complexification gives a subfield of $\mathbf{No}[i]$. The generalized-series interpretation of surreal numbers is part of the standard background reviewed in [5] and in the supplied repository.

Given finitely many actual surreal or surcomplex inputs, take the union of their normal-form supports and the rational span of those exponents, enlarging by 1 when necessary. This is a set-sized divisible ordered subgroup of $\mathbf{No}$. The associated Hahn field contains every input. Finite field operations and the support evaluations above remain in that field. If another workspace contains the same data, uniqueness of the constructed solutions makes the two answers agree in a common enlargement.

Corollary 2.3 (Actual surreal and surcomplex data). *Theorem 1.1, the sharpness assertions, and all subsequent finite precision statements apply to finite configurations in $\mathbf{No}$ or $\mathbf{No}[i]$, using $v(\omega^{-\gamma}) = \gamma$. Every construction is carried out in a set-sized workspace. No quantification over a proper-class coefficient space or use of a proper-class compactness theorem is required.*

The assertion concerns natural-valuation precision of finite algebraic data, not integration, a global surreal exponential, or ordinary paths in the fine topology.

3 Finite moment algebra and the cofactor coordinates

Define

$$P(X) = \prod_{i=1}^n (X - a_i), \quad Q_i(X) = \frac{P(X)}{X - a_i}, \quad p_i = Q_i(a_i) = P'(a_i), \quad c_i = w_i p_i^2. \quad (6)$$

Then $v(p_i) = d_i$ and $v(c_i) = E_i$. Both P and every Q_i are monic integral polynomials. The Q_i form a basis of $K[X]_{<n}$, since evaluation at a_j gives $Q_i(a_j) = \delta_{ij} p_i$.

Let $L : K[X]_{<2n} \rightarrow K$ be the linear functional with $L(X^k) = m_k$. The moment representation is equivalent to

$$L(f) = \sum_i w_i f(a_i), \quad \deg f < 2n.$$

In particular,

$$L(Pq) = 0 \quad (\deg q < n), \quad L(Q_i Q_j) = \delta_{ij} c_i. \quad (7)$$

Proposition 3.1 (Classical Prony uniqueness). *Let $H = (m_{r+s})_{0 \leq r, s < n}$. Then*

$$H = V \operatorname{diag}(w_i) V^\top, \quad \det H = \left(\prod_i w_i \right) \prod_{i < j} (a_j - a_i)^2 \neq 0, \quad (8)$$

where $V_{ri} = a_i^r$. A regular n -node moment vector determines P uniquely by its n orthogonality equations, then the nodes up to permutation, and then the weights. It cannot have a realization with fewer than n nodes.

Proof. The displayed matrix factorization follows entry by entry. The determinant uses the Vandermonde determinant. Writing $P = X^n + \sum_{j < n} P_j X^j$, orthogonality against $1, X, \dots, X^{n-1}$ is the linear system

$$H(P_0, \dots, P_{n-1})^\top = -(m_n, \dots, m_{2n-1})^\top.$$

Any other regular n -node representation must give the same monic annihilator and hence the same nodes. The first n moments then determine the weights through V . A configuration with fewer than n nodes would factor H through a vector space of smaller dimension. $\square$

Since P is monic, (8) reads $\det H = (\prod_i w_i) \text{Disc } P$. With unit weights H is the Hankel matrix of the Newton power sums of P ; this classical identity is the only point of contact with the Hankel section of the collection's spectral-theory report (Section 1.4).

3.1 The structured correction system

Let $\Delta L(X^k) = \epsilon_k$, and $\hat{L} = L + \Delta L$. Seek a monic annihilator in the cofactor coordinates

$$\hat{P} = P + \sum_j b_j Q_j. \quad (9)$$

Testing $\hat{L}(\hat{P}q) = 0$ on $q = Q_i$ yields

$$c_i b_i + \sum_j \Delta L(Q_i Q_j) b_j = -\Delta L(P Q_i).$$

Set

$$u_i = c_i b_i, \quad B_{ij} = \frac{\Delta L(Q_i Q_j)}{c_j}, \quad g_i = \Delta L(P Q_i). \quad (\mathbf{I} + B)u = -g. \quad (10)$$

This identity is exact, not a first-order approximation. In the Q_i basis the perturbed Hankel bilinear form is $(\mathbf{I} + B) \text{diag}(c_i)$. Thus invertibility of $\mathbf{I} + B$ also proves invertibility of the perturbed Hankel matrix.

Lemma 3.2 (Uniform estimate in cofactor coordinates). *If $v(\epsilon_k) \geq \kappa$ and $\kappa > E_j$ for every j , system (10) has a unique solution and*

$$v(u_i) \geq \kappa, \quad v(b_i) \geq \kappa - E_i. \quad (11)$$

Proof. Integrality of $Q_i Q_j$ and $P Q_i$ gives $v(B_{ij}) \geq \kappa - E_j > 0$ and $v(g_i) \geq \kappa$. The matrix $\mathbf{I} + B$ has residue identity, so its inverse has entries in $\mathcal{O}$, by the adjugate formula. Multiplication by that inverse proves the bounds. Alternatively its Neumann series is strongly summable by Lemma 2.1; either proof is valid at arbitrary rank. $\square$

4 A root-localization lemma adapted to the moment problem

Lemma 4.1 (Cofactor localization). *Suppose $v(b_i) > \delta_i$ for every i , and define $\hat{P}$ by (9). Then $\hat{P}$ has one and only one root $\hat{a}_i$ in $B_i = \{x : v(x - a_i) > \delta_i\}$, for every i . These are all its roots, and they are simple. Writing $z_i = \hat{a}_i - a_i$, one has*

$$z_i = -b_i \left(1 + \sum_{j \neq i} \frac{b_j}{a_i - a_j + z_i} \right)^{-1}. \quad (12)$$

Thus $z_i = 0$ precisely when $b_i = 0$, and otherwise $v(z_i) = v(b_i)$ and $z_i/b_i \in -1 + \mathfrak{m}$.

Proof. For $X = a_i + z$ in B_i , none of the factors of $Q_i(X)$ vanishes. Dividing $\hat{P}(X)$ by $Q_i(X)$ gives

$$z + b_i + \sum_{j \neq i} b_j \frac{z}{a_i - a_j + z} = 0. \quad (13)$$

Put $\tau = t^{\delta_i}$, $z = \tau y$, $D_{ij} = a_i - a_j$. The quantities τ/D_{ij} are integral and b_j/D_{ij} are infinitesimal, since $\delta_j \geq d_{ij}$. Multiplying the scaled equation by $\prod_{j \neq i} (1 + (\tau/D_{ij})y)$ gives an integral polynomial whose residue has value zero and derivative one at $y = 0$. By Lemma 2.2 it has exactly one root in $\mathfrak{m}$. Its denominators are units there, so it solves (13).

The balls B_i are disjoint: if a point belonged to B_i and B_j , the ultrametric inequality would give $d_{ij} > \min(\delta_i, \delta_j) \geq d_{ij}$. Consequently the n constructed roots are distinct. A monic polynomial of degree n with n distinct roots has precisely those roots, all simple. Every summand in the parentheses in (12) has positive valuation. The remaining assertions follow from (13). $\square$

For later use, the lemma also gives

$$\frac{\hat{a}_i - \hat{a}_j}{a_i - a_j} \in 1 + \mathfrak{m} \quad (i \neq j). \quad (14)$$

Indeed both node displacements have valuation strictly greater than d_{ij} . In particular, both the valuation and the complex leading coefficient of each difference are unchanged.

5 Reconstruction and the weight estimate

The node estimate alone is not the entire inverse problem. Estimating the last Vandermonde solve as an unrelated operation generally loses the correlations between the polynomial and moment errors. A cross-numerator identity retains precisely those correlations.

5.1 Recovering the numerator and all measured moments

For a formal Laurent series in the independent variable X^{-1} , write $[\cdot]_+$ for its polynomial part. This operation involves only finitely many coefficients in the following formulas. Set

$$A = \sum_i w_i Q_i, \quad \hat{A} = \left[\hat{P}(X) \sum_{k=0}^{2n-1} \hat{m}_k X^{-k-1} \right]_+. \quad (15)$$

Both numerators have degree less than n .

Lemma 5.1 (Finite Padé identity). *If $\hat{L}(\hat{P}q) = 0$ for every $\deg q < n$, then*

$$\frac{\hat{A}(X)}{\hat{P}(X)} = \sum_{k=0}^{2n-1} \hat{m}_k X^{-k-1} + O_X(X^{-2n-1}). \quad (16)$$

When $\hat{P}$ has distinct roots $\hat{a}_i$, its residues $\hat{w}_i = \hat{A}(\hat{a}_i)/\hat{P}'(\hat{a}_i)$ realize all $2n$ moments. Here O_X refers only to formal degree in X^{-1} , not to a valuation limit.

Proof. In the product used in (15), the coefficient of X^{-r-1} , for $0 \leq r < n$, is $\sum_{j=0}^n \hat{P}_j \hat{m}_{j+r} = \hat{L}(X^r \hat{P})$, hence zero. After subtracting the polynomial part, the product therefore begins at X^{-n-1} . Division by the monic degree- n polynomial produces (16). Partial fractions give $\hat{A}/\hat{P} = \sum_i \hat{w}_i/(X - \hat{a}_i)$, and expanding each fraction at infinity identifies its first $2n$ coefficients. $\square$

Lemma 5.2 (Cross-numerator control). *Suppose $P, \hat{P}$ are monic integral degree- n polynomials and their associated rational functions agree with the original and perturbed moment vectors through X^{-2n} . Then*

$$D = \hat{A}P - A\hat{P}, \quad \deg D < 2n, \quad (17)$$

has every coefficient of valuation at least κ whenever $v(\epsilon_k) \geq \kappa$. In the cofactor coordinates,

$$D(a_i) = -w_i p_i^2 b_i = -u_i. \quad (18)$$

Proof. Multiply the Laurent expansion of $\hat{A}/\hat{P} - A/P$ by $P\hat{P}$ and take its polynomial part. The tail $O_X(X^{-2n-1})$ contributes no polynomial term. Every coefficient of D is consequently a finite sum of moment errors multiplied by integral coefficients of $P\hat{P}$. This proves the bound. At a_i , use $A(a_i) = w_i p_i$ and $\hat{P}(a_i) = b_i p_i$. $\square$

5.2 A divided-difference estimate for the weights

Lemma 5.3 (Weight errors without a second conditioning loss). *Under the hypotheses of Lemma 5.2, suppose the roots are labelled by $v(\hat{a}_i - a_i) > \delta_i$. Then*

$$v(\hat{w}_i - w_i) \geq \kappa - 2d_i - \delta_i.$$

Proof. Fix i , write $a = a_i$, $b = \hat{a}_i$, and put

$$q(X) = Q_i(X) \frac{\hat{P}(X)}{X - \hat{a}_i}, \quad F(X) = \frac{D(X)}{q(X)}.$$

All factors of q are nonzero at a and b . Preservation of pairwise separation gives

$$v(q(a)) = v(q(b)) = 2d_i. \quad (19)$$

If $a \neq b$, the residues at a and b in

$$\frac{\hat{A}}{\hat{P}} - \frac{A}{P} = \frac{F(X)}{(X - a)(X - b)}$$

are $-w_i$ and $\hat{w}_i$. Thus

$$\hat{w}_i - w_i = \frac{F(b) - F(a)}{b - a}. \quad (20)$$

If $a = b$, the left side has at most a simple pole at a , so $D(a) = 0$, and the residue gives $\hat{w}_i - w_i = F'(a)$. This is the coincident-argument version of the same formula.

For distinct arguments, denote a divided difference by $[a, b]f$. Because $a, b \in \mathcal{O}$ and every coefficient of D has valuation at least κ , expansion of $[a, b]X^k$ as a finite geometric sum gives $v([a, b]D) \geq \kappa$. The same holds for $D'(a)$. Each of the $2n - 2$ linear factors of q has value of valuation d_{ij} at both arguments, with each $j \neq i$ occurring twice. Telescoping the product difference, or differentiating in the coincident case, therefore shows

$$v([a, b]q) \geq 2d_i - \delta_i.$$

For $n = 1$, $q = 1$ and this divided difference is zero, which also satisfies the bound. Finally the quotient identity

$$[a, b]F = \frac{[a, b]D}{q(b)} - \frac{D(a)[a, b]q}{q(a)q(b)}$$

shows that its valuation is at least $\min(\kappa - 2d_i, \kappa - 2d_i - \delta_i) = \kappa - 2d_i - \delta_i$. The derivative quotient formula proves the same estimate when $a = b$. $\square$

5.3 Proof of the reconstruction assertions

Proof of Theorem 1.1, except sharpness. Since $\delta_i \geq 0$ and $\kappa > \Theta$, Lemma 3.2 constructs the unique cofactor correction with $v(b_i) \geq \kappa - E_i > \delta_i$. Lemma 4.1 constructs the n simple roots, establishes the labelling, and proves (4). They are integral, so their product $\hat{P}$ is integral.

Use Lemma 5.1 to recover the weights and all measured moments. Lemmas 5.2 and 5.3 give (5). Since this bound is strictly greater than $v(w_i) = \varpi_i$, the new weights are nonzero and $\hat{w}_i/w_i \in 1 + \mathfrak{m}$. Thus the constructed realization is regular. The perturbed Hankel matrix is invertible by (10); alternatively, its factorization through these distinct nodes and nonzero weights proves the same fact. Proposition 3.1 gives uniqueness among all regular n -node configurations. The balls are disjoint, so their labelling is unique. Formula (14) supplies the final leading-term assertion for the node differences. $\square$

Corollary 5.4 (Real and positive data). *For real Hahn or actual surreal inputs, the recovered nodes and weights in Theorem 1.1 are real. Every initially positive weight remains positive. The ordering of the nodes is preserved.*

Proof. The construction takes place over the real coefficient field. Alternatively, complex conjugation fixes the data and the labelled balls, so uniqueness fixes every output. The ratios of corresponding weights and pairwise node differences lie in $1 + \mathfrak{m}$, which consists of positive real Hahn elements. $\square$

Remark 5.5 (What the theorem does not require). Weights need not be finite, positive, or have distinct leading coefficients. Their valuations may be negative. Nodes may have identical standard parts, and their mutual distances may have different Archimedean orders. No assumption on the discriminant being an ordinary nonzero number is made. The main proof uses only finite valued-field algebra and residue-simple lifting; the Hahn setting additionally supplies explicit support control.

6 An exact worst direction and sharpness at every configuration

The same perturbation direction works for every configuration: change only the highest measured moment. It is the familiar last-coordinate Prony-curve direction [3]; the point here is its exact threshold consequence.

Proposition 6.1 (The last-moment formula). *Let $\epsilon_{2n-1} = e$ and all other $\epsilon_k = 0$. The original Hankel matrix is unchanged. The unique monic annihilator determined by the new moments is*

$$P_e(X) = P(X) - e \sum_{i=1}^n \frac{Q_i(X)}{w_i P'(a_i)^2}. \quad (21)$$

This formula is exact for every $e \in K$, regardless of its valuation.

Proof. Every $Q_i Q_j$ has degree at most $2n - 2$, so $B = 0$ in (10). Each PQ_i is monic of degree $2n - 1$, so $g_i = e$. Thus $u_i = -e$ and $b_i = -e/c_i$. The unchanged Hankel matrix is invertible, which proves uniqueness without any smallness hypothesis. $\square$

Proof of the sharpness assertions in Theorem 1.1. If $v(e) = \kappa > \Theta$, then $v(b_i) = \kappa - E_i > \delta_i$. Lemma 4.1 yields

$$v(\hat{a}_i - a_i) = \kappa - E_i, \quad \frac{\hat{a}_i - a_i}{e/(w_i P'(a_i)^2)} \in 1 + \mathfrak{m}. \quad (22)$$

This proves simultaneous attainment of all the node bounds.

Now take $v(e) = \Theta$ and choose i with $E_i + \delta_i = \Theta$. Any regular n -node realization must have annihilator P_e , by Proposition 3.1. Formula (21) gives

$$v(P_e(a_i)) = v(b_i p_i) = \Theta - E_i + d_i = \delta_i + d_i.$$

If all its nodes admitted the strict-ball labelling, factorization at a_i would instead give

$$v(P_e(a_i)) = v(a_i - \hat{a}_i) + \sum_{j \neq i} v(a_i - \hat{a}_j) > \delta_i + d_i.$$

The other factors have precisely their original separation valuations by the strict-ball assumption. This is a contradiction. $\square$

Because this boundary perturbation is allowed in every error ball with precision $\kappa \leq \Theta$, the universal strict-ball reconstruction guarantee holds *if and only if* $\kappa > \Theta$.

6.1 Cluster interpretation and comparison with a determinant

When every weight has valuation zero and all n nodes occupy one equally spaced valuation cluster, $d_{ij} = s > 0$, one has

$$d_i = (n - 1)s, \quad \Theta = (2n - 1)s.$$

The pointwise geometry matters when this model is not uniform. A node deep inside a nested cluster has one separation contribution for each other node, not merely $n - 1$ copies of the nearest separation. The determinant records only a sum:

$$v(\det H) = \sum_i \varpi_i + 2 \sum_{i < j} d_{ij} = \sum_i \varpi_i + \sum_i d_i. \quad (23)$$

This single number does not specify the vector $(E_i + \delta_i)_i$ and does not replace the threshold or the individual error bounds.

6.2 Comparison with the coefficient threshold

This subsection was added when the manuscript joined the collection. It compares Theorem 1.1 with the coefficient-perturbation theorem `polynomial:thm:stability` of the polynomial-algebra report [8]. It proves no new theorem; every step is an elementary consequence of results stated above or in that report.

Remark 6.2 (Moment input versus coefficient input). For $P = \prod_i (X - a_i)$, the root data of `polynomial:thm:stability` are the d_{ij} , d_i and δ_i of (2), and its threshold is $T(P) = \max_i (d_i + \delta_i)$. By (2),

$$\Theta = \max_i ((\varpi_i + d_i) + (d_i + \delta_i)).$$

The extra summand $\varpi_i + d_i = v(w_i P'(a_i))$ is the cost, at a_i , of passing from moments to the annihilator. Indeed, the perturbed annihilator (9) differs from P by $\sum_j b_j Q_j$, a polynomial of degree less than n whose value at a_i is $b_i p_i$; every polynomial of degree less than n has this form, with b_i its value at a_i divided by p_i . Lemma 3.2 gives $v(b_i p_i) \geq \kappa - \varpi_i - d_i$, and Lemma 4.1 needs $v(b_i p_i) > d_i + \delta_i$, the i th summand of $T(P)$, at every node. Replacing the coefficient precision ε of

`polynomial:thm:stability` by the nodewise annihilator precision $\kappa - \varpi_i - d_i$ turns its displacement bound $\varepsilon - d_i$ into the node bound $\kappa - E_i$ of (4). If every $\varpi_i \geq 0$, then $\Theta \geq T(P)$. Weights of negative valuation can make Θ smaller: two nodes at separation valuation $s > 0$ with both weights t^{-2s} give $\Theta = s < 2s = T(P)$.

Three further points make the comparison precise.

- (a) Lemma 4.1 is a nodewise form of the matching step of `polynomial:thm:stability`. If every coefficient of $\hat{P} - P$ has valuation greater than $T(P)$, then its values at the integral nodes satisfy $v(b_i p_i) > d_i + \delta_i$, which is the hypothesis of the lemma. The converse fails. For the nodes $0, t^s, 1$ with $s > 0$ one has $T(P) = 2s$ and $\delta_3 = 0$, so $\hat{P} = P + t^s Q_3$ satisfies the hypothesis of the lemma, while $\hat{P} - P = t^s X^2 - t^{2s} X$ has a coefficient of valuation $s < T(P)$. The lemma yields the matching, the uniqueness of the labelling, the preserved separations (14) and the exact displacement valuation $v(\hat{a}_i - a_i) = v(b_i)$. It does not restate the second-order Newton estimate or the discriminant statement of `polynomial:thm:stability`.
- (b) Theorem 1.1 does not follow merely by composing Lemma 3.2 with `polynomial:thm:stability`, which needs every coefficient of $\hat{P} - P$ to have valuation above $T(P)$. For the two nodes $0, h = t^s$ with unit weights, the last-moment perturbation by e with $v(e) = \kappa$, formula (39), gives $\hat{P} - P = -(2e/h^2)X + e/h$, with a coefficient of valuation exactly $\kappa - 2s$, while $T(P) = 2s$. For $3s = \Theta < \kappa \leq 4s$ this perturbation is outside the hypothesis of `polynomial:thm:stability`, whereas Theorem 1.1 applies.
- (c) The sharpness statements differ in scope. `polynomial:ex:sharpdisc` shows, for one family, that the strict inequality in `polynomial:thm:stability` cannot be relaxed. Here the last-moment direction of Proposition 6.1 defeats the strict-ball guarantee at valuation Θ for every configuration. No coefficient-side analogue of this universal statement is claimed.

Finally, $\det H = (\prod_i w_i) \text{Disc } P$, so (23) reads $v(\det H) = \sum_i \varpi_i + D$, where $D = v(\text{Disc } P) = \sum_i d_i$ is the discriminant valuation of `polynomial:cor:discprecision`.

6.3 Infinite scales and affine normalization

The integrality assumption on nodes is a normalization, not a restriction on finite configurations in $\mathbf{No}[i]$. Choose $c, \rho \in K$, $\rho \neq 0$, with $x_i = (a_i - c)/\rho \in \mathcal{O}$. The normalized moments are

$$m_k^{\text{norm}} = \rho^{-k} \sum_{\ell=0}^k \binom{k}{\ell} (-c)^{k-\ell} m_\ell. \quad (24)$$

The same linear transformation must be applied to the errors. For error bounds $v(\epsilon_\ell) \geq \kappa_\ell$, a valid normalized bound is

$$v(\epsilon_k^{\text{norm}}) \geq \min_{0 \leq \ell \leq k} (\kappa_\ell + (k - \ell)v(c) - kv(\rho)),$$

with zero coefficients omitted; for $c = 0$ only $\ell = k$ contributes. Apply the uniform theorem to a common lower bound, or the nonuniform certificate below to the individual normalized bounds. Translating back adds $v(\rho)$ to each node-error valuation and does not change the weights. One must not rescale the nodes while leaving the moment-error model unchanged.

7 Exact local precision and cancellation in the Hermite inverse

7.1 The differential and its classical dual basis

Order the coordinates of the parameter space as $(w_1, \dots, w_n, a_1, \dots, a_n)$, and write $\mathcal{M} : K^{2n} \rightarrow K^{2n}$ for the moment map (1). At a regular configuration its differential is

$$dm_k = \sum_i \left(a_i^k dw_i + w_i k a_i^{k-1} da_i \right). \quad (25)$$

The derivative term at $k = 0$ is zero, including when $a_i = 0$. Define

$$\ell_i(X) = Q_i(X)/p_i, \quad \chi_i = \ell'_i(a_i) = \sum_{j \neq i} \frac{1}{a_i - a_j}, \quad (26)$$

$$H_i(X) = (1 - 2\chi_i(X - a_i))\ell_i(X)^2, \quad G_i(X) = (X - a_i)\ell_i(X)^2.$$

These are the usual degree- $< 2n$ Hermite interpolation polynomials. They satisfy

$$H_i(a_j) = \delta_{ij}, \quad H'_i(a_j) = 0, \quad G_i(a_j) = 0, \quad G'_i(a_j) = \delta_{ij}.$$

Consequently the inverse differential is

$$dw_i = (dL)(H_i), \quad da_i = (dL)(G_i)/w_i. \quad (27)$$

This proves that the Jacobian J is invertible and identifies each row of J^{-1} . These interpolation identities are classical; the precision calculation below specifies exactly what they imply over Hahn coefficients.

For a nonzero polynomial f , let $v_G(f) = \min_k v([X^k]f)$. This Gauss valuation is multiplicative: normalize each factor by a monomial to make its minimum zero and reduce modulo $\mathfrak{m}$; the product of two nonzero residue polynomials is nonzero. Define the cancellation depth

$$q_i = \begin{cases} \max(0, -v(\chi_i)), & \chi_i \neq 0, \\ 0, & \chi_i = 0. \end{cases} \quad 0 \leq q_i \leq \delta_i. \quad (28)$$

The inequality follows from the ultrametric inequality for the sum defining χ_i .

Proposition 7.1 (Exact row losses). *For finite nodes,*

$$v_G(G_i) = -2d_i, \quad v_G(H_i) = -2d_i - q_i. \quad (29)$$

Thus the exact coordinate projections of the tangent precision lattice $J^{-1}t^\kappa \mathcal{O}^{2n}$ are

$$da_i \in t^{\kappa - E_i} \mathcal{O}, \quad dw_i \in t^{\kappa - 2d_i - q_i} \mathcal{O}. \quad (30)$$

Each displayed projection is the whole indicated fractional ideal, not merely a containing ball.

Proof. The monic integral polynomials Q_i and $X - a_i$ have Gauss valuation zero, so $v_G(\ell_i^2) = -2d_i$ and the first formula follows. If $\chi_i = 0$, the remaining linear factor in H_i is one. If $v(\chi_i) < 0$, its coefficient of X has valuation $v(\chi_i)$ and its constant coefficient has at least that valuation. If $v(\chi_i) \geq 0$, the factor is integral and evaluates to one at the integral point a_i , so its Gauss valuation is zero. Here $v(2) = 0$ since the coefficient field is $\mathbb{R}$ or $\mathbb{C}$. Multiplicativity gives the second formula.

The image of an $\mathcal{O}$ -linear row is the fractional ideal generated by its finitely many coefficients. Such an ideal is generated by a coefficient of minimum valuation. Equations (27) and (29) now prove (30), including surjectivity onto each ideal separately. $\square$

7.2 A support-based exact inverse-lattice theorem

The following proposition is an arbitrary-rank polynomial implementation of the differential precision principle in [4]. It is included with proof so that no Banach-space convergence hypothesis is silently imported.

Proposition 7.2 (Finite coefficient criterion). *Let $F : K^N \rightarrow K^N$ be polynomial, let $x \in K^N$, and suppose $J = dF_x$ is invertible. Expand, in independent variables Z ,*

$$F(x + J^{-1}Z) - F(x) = Z + \left(\sum_{|\nu| \geq 2} C_{j,\nu} Z^\nu \right)_{j=1}^N. \quad (31)$$

The sums are finite. If $\kappa \in \Gamma$ satisfies

$$v(C_{j,\nu}) + (|\nu| - 1)\kappa > 0 \quad \text{for every nonzero } C_{j,\nu}, \quad (32)$$

then F is a bijection from $x + J^{-1}t^\kappa \mathcal{O}^N$ onto $F(x) + t^\kappa \mathcal{O}^N$. In particular,

$$F(x + J^{-1}t^\kappa \mathcal{O}^N) = F(x) + t^\kappa \mathcal{O}^N. \quad (33)$$

Proof. Scale $Z = t^\kappa Y$ and divide the output increment by t^κ . The normalized map is $T(Y) = Y + R(Y)$, where every coefficient of R belongs to $\mathfrak{m}$ and every monomial has degree at least two. For $Y, Y' \in \mathcal{O}^N$, polynomial divided differences give

$$R(Y) - R(Y') = A(Y, Y')(Y - Y')$$

with every entry of $A(Y, Y')$ in $\mathfrak{m}$. If $Y \neq Y'$, the minimum valuation of this correction is strictly greater than $v_{\min}(Y - Y')$. It follows that

$$v_{\min}(T(Y) - T(Y')) = v_{\min}(Y - Y'),$$

which proves injectivity.

For a prescribed $b \in \mathcal{O}^N$, the polynomial system $T(Y) = b$ reduces to $Y = \bar{b}$ modulo $\mathfrak{m}$, with residue Jacobian identity. Lemma 2.2 gives a solution in $\bar{b} + \mathfrak{m}^N \subset \mathcal{O}^N$. This proves surjectivity. The lemma's proof evaluates a formal series in finitely many positive-support coefficients and in $b - \bar{b}$; no assertion about an ordinary Cauchy sequence is involved. Undo the two linear changes of variables. $\square$

Corollary 7.3 (Exact nonlinear uncertainty set). *For the Prony map, assume $\kappa > \Theta$ and also (32) for its finite expansion (31). Then the set of all parameter increments with strict-ball labelling realizing moment errors in $t^\kappa \mathcal{O}^{2n}$ is exactly*

$$J^{-1}t^\kappa \mathcal{O}^{2n}. \quad (34)$$

In particular the exact coordinate ranges are

$$\hat{a}_i - a_i \in t^{\kappa - E_i} \mathcal{O}, \quad \hat{w}_i - w_i \in t^{\kappa - 2d_i - q_i} \mathcal{O}.$$

Proof. By Proposition 7.1, every point of the proposed parameter lattice satisfies strict node matching and preserves nonzero weights: $q_i \leq \delta_i$ and $\kappa > \Theta$ give the required inequalities. Proposition 7.2 proves exact surjectivity onto the error ball, and Theorem 1.1 gives uniqueness of its labelled realization. The coordinate projections were computed exactly in Proposition 7.1. $\square$

The lattice in (34) is generally not the Cartesian product of its coordinate projections: errors in different recovered parameters are correlated. Since $\mathbb{k}$ is infinite, all finite row minima can nevertheless be attained simultaneously. Normalize each nonzero row of J^{-1} by its least valuation and choose an ordinary vector avoiding the finitely many resulting residue hyperplanes. Its lattice image has the minimum valuation in every parameter coordinate, and its actual moment increment belongs to the prescribed error ball.

There always exists a sufficiently large κ satisfying the finite criterion: choose it positive and larger than every $-v(C_{j,\nu})$ and than Θ . In a divisible group one may use the sharper sufficient inequalities $\kappa > -v(C_{j,\nu})/(|\nu| - 1)$. This auxiliary criterion is *not* claimed to be the optimal onset of exact lattice behavior. It does not weaken the sharp threshold theorem, which requires only $\kappa > \Theta$ and was proved independently.

8 Nonuniform moment precision and a directed-path certificate

Uniform measurement precision is convenient but sometimes wasteful. For this section assume Γ is divisible, as it may always be chosen for actual surreal data. Suppose

$$v(\epsilon_k) \geq \kappa_k \in \Gamma, \quad 0 \leq k < 2n.$$

For a nonzero polynomial of degree less than $2n$, define its noise gauge

$$\nu_\kappa(f) = \min_{[X^k]f \neq 0} (\kappa_k + v([X^k]f)). \quad (35)$$

This always bounds $v(\Delta L(f))$ from below, although cancellations may make the actual value higher. Put

$$\lambda_i = \nu_\kappa(PQ_i), \quad \mu_{ij} = \nu_\kappa(Q_iQ_j), \quad g_{ij} = \mu_{ij} - E_j. \quad (36)$$

Make a directed graph with vertices $1, \dots, n$; the edge from j to i has weight g_{ij} . Loops are included. A path starting at j has initial cost λ_j , plus the sum of its edge weights.

Theorem 8.1 (Nonuniform reconstruction certificate). *Suppose every directed simple cycle, including every loop, has strictly positive total edge weight. For each i , let ρ_i be the minimum cost of a simple path ending at i , with length-zero paths allowed. If*

$$\rho_i - E_i > \delta_i \quad \text{for every } i, \quad (37)$$

then every allowed moment perturbation has a unique regular n -node realization. It admits strict-ball labelling and satisfies

$$v(\hat{a}_i - a_i) \geq \rho_i - E_i. \quad (38)$$

If $\kappa_ = \min_k \kappa_k$, a further, possibly weak, weight bound is $v(\hat{w}_i - w_i) \geq \kappa_* - 2d_i - \delta_i$. The certificate is sufficient; no necessity or universal sharpness is claimed for these graph conditions.*

Proof. Choose $\eta > 0$ less than every simple cycle's mean edge weight. There are only finitely many simple cycles, and divisibility of Γ permits this choice. Replace each edge weight by $g_{ij} - \eta$. Every directed cycle now has positive weight, because a closed walk decomposes into simple cycles. Add a source with a zero-weight edge to every vertex, and let ζ_i be its shortest-path cost to i . Shortest costs are attained by simple paths, since deleting cycles decreases cost. They satisfy

$$\zeta_i \leq \zeta_j + g_{ij} - \eta, \quad g_{ij} + \zeta_j - \zeta_i \geq \eta.$$

Let $T = \text{diag}(t^{\zeta_i})$. The matrix $B' = T^{-1}BT$ has every entry of positive valuation, by (10) and (36). Hence $I + B'$ and $I + B$ are invertible. Moreover the Neumann expansion is strongly summable, by Lemma 2.1 applied to the finitely many entries of B' . Multiplying by a fixed finite vector of Hahn series preserves this strong summability.

In

$$u = - \sum_{\ell \geq 0} (-B)^\ell g,$$

every matrix-product term is indexed by a directed walk ending at its output vertex. Its valuation is at least the initial λ plus its edge weights. Deleting repeated-vertex cycles only decreases this lower bound. Thus every such term has valuation at least ρ_i , and so $v(u_i) \geq \rho_i$. Strong summability legitimizes the expansion and regrouping even if no multiple of η is cofinal in Γ . It follows that $v(b_i) \geq \rho_i - E_i > \delta_i$.

Construct the n roots by Lemma 4.1 and the weights by Lemma 5.1. These weights are nonzero even when the coarse weight bound is inconclusive: the perturbed Hankel matrix is invertible because its matrix in the Q_i basis is $(I + B) \text{diag}(c_i)$, whereas its factorization through the constructed distinct nodes would be singular if any weight vanished. Classical Prony uniqueness completes the reconstruction assertion. The root valuation follows from the same lemma; Lemma 5.3 with κ_* gives the last bound. $\square$

8.1 How to compute the certificate

The certificate requires finite polynomial multiplication, finitely many coefficient valuations, and ordered-group arithmetic. One may enumerate simple cycles and paths for a small configuration. More efficiently, shortest-path relaxation supplies the potentials and the costs once a valid positive margin is chosen. The value group need not be represented by real numbers: rational tuples with lexicographic comparison work directly. The polynomial coefficients themselves must be known well enough to decide zero coefficients and their valuations.

The gauge (35) discards some information: the quantities $\Delta L(PQ_i)$ and $\Delta L(Q_iQ_j)$ are built from the *same* errors. Cancellations and cross-entry relations can make the graph certificate fail while reconstruction is still well behaved. It would be incorrect to advertise this sufficient method as the exact admissible precision region. In the uniform case, taking $\lambda_i \geq \kappa$ and $g_{ij} \geq \kappa - E_j > 0$ immediately recovers the bounds used in Theorem 1.1.

9 Worked examples

9.1 One atom and the meaning of the empty maximum

For $n = 1$, $m_0 = w$ and $m_1 = wa$. The exact perturbed parameters are

$$\hat{w} = w + \epsilon_0, \quad \hat{a} = \frac{wa + \epsilon_1}{w + \epsilon_0}, \quad \hat{a} - a = \frac{\epsilon_1 - a\epsilon_0}{w + \epsilon_0}.$$

Our convention gives $d = \delta = 0$, $E = \Theta = \varpi = v(w)$. Errors of valuation at least $\kappa > \varpi$ preserve the weight's leading term and put $\hat{a}$ in $a + \mathfrak{m}$, with node valuation bound $\kappa - \varpi$ and weight bound κ . Taking $\epsilon_0 = 0$, $\epsilon_1 = e$ attains the node bound; at $v(e) = \varpi$, the new node is not in $a + \mathfrak{m}$. This example checks the degree-one endpoint conventions directly.

9.2 Two nodes: exact formulas and an actual boundary collision

Let $a_1 = 0$, $a_2 = h = t^s$, $s > 0$, and $w_1 = w_2 = 1$. Then

$$(m_0, m_1, m_2, m_3) = (2, h, h^2, h^3), \quad d_i = \delta_i = s, \quad E_i = 2s, \quad \Theta = 3s.$$

Perturb only m_3 by e . Proposition 6.1 gives

$$P_e(X) = X^2 - \left(h + \frac{2e}{h^2}\right)X + \frac{e}{h}, \quad \text{Disc}(P_e) = h^2 + \frac{4e^2}{h^4}. \quad (39)$$

For $e = h^4$, put $S = \sqrt{1 + 4h^2}$, the root in $1 + \mathfrak{m}$. The strict-ball-labelled nodes and weights are

$$\begin{aligned} \hat{a}_1 &= (h + 2h^2 - hS)/2, & \hat{w}_1 &= 1 + 2h/S, \\ \hat{a}_2 &= (h + 2h^2 + hS)/2, & \hat{w}_2 &= 1 - 2h/S. \end{aligned}$$

Their initial terms are

$$\begin{aligned} \hat{a}_1 &= h^2 - h^3 + h^5 - 2h^7 + O_h(h^9), \\ \hat{a}_2 &= h + h^2 + h^3 - h^5 + 2h^7 + O_h(h^9), \\ \hat{w}_1 &= 1 + 2h - 4h^3 + 12h^5 - 40h^7 + O_h(h^9), \\ \hat{w}_2 &= 1 - 2h + 4h^3 - 12h^5 + 40h^7 + O_h(h^9). \end{aligned}$$

Here $O_h(h^9)$ denotes formal power-series order, hence Hahn valuation at least 9s. Both node displacements have valuation $2s = 4s - E_i$; both weight errors have valuation $s = 4s - 2d_i - \delta_i$.

At the boundary take $e = ch^3$ with ordinary $c \neq 0$. Formula (39) becomes

$$P_e(X) = X^2 - h(1 + 2c)X + ch^2.$$

For $c = i/2$, the discriminant is zero and the root is $h(1 + i)/2$. The unchanged Hankel determinant is $h^2 \neq 0$, so no regular two-node realization exists. For nonzero real c , the roots instead are

$$\frac{h}{2} \left(1 + 2c \pm \sqrt{1 + 4c^2}\right),$$

with positive weights $1 \pm 2c/\sqrt{1 + 4c^2}$ assigned in the reverse sign order. They give a regular positive real realization, but neither node retains its strict nearest-neighbour label. Thus failure at the threshold need not mean a collision or a loss of positivity.

9.3 The same tree, different weight precision

Let the three nodes be $-h, 0, h$, with $h = t^s$, $s > 0$, and all weights one. Every separation valuation is s , so

$$d_i = 2s, \quad \delta_i = s, \quad E_i = 4s, \quad \Theta = 5s.$$

At the middle node,

$$\chi_2 = \frac{1}{h} + \frac{1}{-h} = 0,$$

whereas at the two endpoints $\chi_1 = -3/(2h)$ and $\chi_3 = 3/(2h)$. Hence $q_2 = 0$ and $q_1 = q_3 = s$. The exact weight row losses are consequently $4s$ at the middle and $5s$ at the endpoints. Under the

explicit extra smallness criterion of Corollary 7.3, the middle weight's full nonlinear uncertainty range is $t^{\kappa-4s}\mathcal{O}$, a factor t^s smaller than the endpoint ranges $t^{\kappa-5s}\mathcal{O}$.

For comparison take the nodes $0, h, 3h$. They have exactly the same labelled separation valuations. Now

$$\chi_1 = -\frac{4}{3h}, \quad \chi_2 = \frac{1}{2h}, \quad \chi_3 = \frac{5}{6h},$$

so all three q_i equal s . Thus the full separation matrix determines the uniform threshold and the node losses, but does *not* determine the exact weight precision. Residue-level cancellation contains additional information. The extra nonlinear lattice criterion is essential to the stated nonlinear interpretation; the tangent calculation needs no such criterion.

Proposition 9.1 (A nonlinear obstruction to the cancellation gain). *For the symmetric configuration $(-h, 0, h)$ with unit weights, perturb only m_5 by e , with $v(e) = \kappa > 5s$. The recovered middle weight satisfies*

$$v(\hat{w}_2 - 1) = 2\kappa - 10s. \quad (40)$$

Consequently, whenever $5s < \kappa < 6s$, its nonlinear error violates the improved tangent bound $\kappa - 4s$. The exact Hermite lattice cannot describe the whole inverse error set throughout the entire sharp-threshold region.

Proof. Normalize the nodes by $X = hY$ and put $\lambda = e/h^5 \in \mathfrak{m}$. The original normalized moments are $(3, 0, 2, 0, 2, 0)$ and the perturbed last moment is λ . The exact monic annihilator and numerator are

$$P_\lambda(Y) = Y^3 - \frac{3}{2}\lambda Y^2 - Y + \lambda, \quad A_\lambda(Y) = 3Y^2 - \frac{9}{2}\lambda Y - 1.$$

These follow from (21) and (15), or directly from the three orthogonality equations. The middle root $y \in \mathfrak{m}$ satisfies

$$y \left(1 - y^2 + \frac{3}{2}\lambda y \right) = \lambda,$$

so $v(y) = v(\lambda) = \kappa - 5s$. Its recovered weight obeys

$$\hat{w}_2 - 1 = \frac{A_\lambda(y) - P'_\lambda(y)}{P'_\lambda(y)} = \frac{-\frac{3}{2}\lambda y}{3y^2 - 3\lambda y - 1}.$$

The denominator is a unit of residue -1 , giving (40). For $\kappa < 6s$, $2\kappa - 10s < \kappa - 4s$, as asserted. Values in this interval exist in the divisible workspaces used for actual surreal numbers. $\square$

For the same symmetric example, the explicit choice $\kappa > 9s$ is a simple sufficient condition for the full nonlinear lattice theorem. Indeed the inverse Hermite lattice gives weight increments of valuation at least $\kappa - 5s$ and node increments of valuation at least $\kappa - 4s$. In the k th moment, a nonlinear term with $j \geq 2$ node increments and no weight increment has valuation at least $j\kappa + (k - 5j)s$, with $k \geq j$. A term with one weight increment and $j \geq 1$ node increments has valuation at least $(j + 1)\kappa + (k - 5j - 5)s$, again with $k \geq j$. For $\kappa > 9s$ both bounds are strictly greater than κ . The zeroth moment is linear. Applying these estimates coefficientwise in the independent normalized moment variables verifies (32). This is a sufficient hand-checkable bound, not an optimal one; the obstruction just proved occurs below $6s$.

9.4 A genuinely higher-rank configuration

Take $\Gamma = \mathbb{Q}e_1 \oplus_{\text{lex}} \mathbb{Q}e_2$, with $e_1 > Ne_2 > 0$ for every positive integer N , and put

$$(a_1, a_2, a_3, a_4) = (0, t^{e_1}, t^{e_2}, 1), \quad (w_1, w_2, w_3, w_4) = (1, t^{e_2}, 1, t^{2e_2}).$$

The only positive separation valuations are $d_{12} = e_1$ and $d_{13} = d_{23} = e_2$. The following table records the full threshold data.

i	ϖ_i	d_i	δ_i	$E_i + \delta_i$
1	0	$e_1 + e_2$	e_1	$3e_1 + 2e_2$
2	e_2	$e_1 + e_2$	e_1	$3e_1 + 3e_2$
3	0	$2e_2$	e_2	$5e_2$
4	$2e_2$	0	0	$2e_2$

Thus $\Theta = 3e_1 + 3e_2$. For $\kappa = 3e_1 + 4e_2$, the guaranteed node precisions are, respectively,

$$e_1 + 2e_2, \quad e_1 + e_2, \quad 3e_1, \quad 3e_1 + 2e_2.$$

The weight-error bounds are

$$2e_2, \quad 2e_2, \quad 3e_1 - e_2, \quad 3e_1 + 4e_2.$$

Each weight-error bound is strictly above the original weight's valuation. Perturbing the seventh moment, m_7 , by t^κ attains all four node bounds exactly. All these assertions are ordered-pair arithmetic; no real-valued logarithm is involved.

In actual surreals one may choose $e_1 = \omega$, $e_2 = 1$. The nodes then are $0, \omega^{-\omega}, \omega^{-1}, 1$, and the admissible test error is $\omega^{-(3\omega+4)}$. There is no ordered-group embedding of this entire Γ into the additive reals: a positive image of e_2 cannot have all its integer multiples below a fixed image of e_1 . This does not deny that a finite collection of these inequalities can sometimes be emulated by a sufficiently large real scale ratio; it explains why one such ratio cannot represent the whole group.

9.5 Nonuniform errors for two nodes

Return to $0, h$ with weights one. For $Q_1 = X - h$, $Q_2 = X$ and $P = X^2 - hX$, the noise gauges are

$$\begin{aligned} \lambda_1 &= \min(\kappa_3, \kappa_2 + s, \kappa_1 + 2s), \\ \lambda_2 &= \min(\kappa_3, \kappa_2 + s), \\ \mu_{11} &= \min(\kappa_2, \kappa_1 + s, \kappa_0 + 2s), \\ \mu_{12} = \mu_{21} &= \min(\kappa_2, \kappa_1 + s), \quad \mu_{22} = \kappa_2. \end{aligned}$$

Suppose

$$\kappa_0 > 0, \quad \kappa_1 > s, \quad \kappa_2 > 2s, \quad \kappa_3 > 3s. \quad (41)$$

Then all graph edges have positive weight, and all initial costs exceed $3s$. Every ρ_i therefore exceeds $3s$, so the certificate applies. These errors need not have a common precision above $3s$.

There is also a simple direct interpretation: normalize $X = hY$ and divide the k th moment by h^k . The normalized nodes are $0, 1$, their threshold is zero, and all normalized errors in (41) are infinitesimal. The main theorem then also proves preservation of the leading weights. This example illustrates the certificate; it is not claimed as a new scaling principle.

10 Constructive content, symbolic checks, and formalization

10.1 A finite-algebra reconstruction procedure

With exact input data and a certified precision hypothesis, the proof gives a specific procedure. Form P, Q_i, p_i, c_i , compute the finite quantities $\Delta L(Q_i Q_j)$ and $\Delta L(P Q_i)$, solve $(I + B)u = -g$, and form $\hat{P} = P + \sum_i (u_i/c_i)Q_i$. For each i , solve the residue-simple local equation (13) by the formal-parameter construction. Finally compute $\hat{A}$ by finite polynomial-part extraction and the weights by residues. All node corrections and their labels are certified before the last step.

This is a mathematical construction, not an unconditional Turing algorithm on arbitrary surreal inputs. An effective implementation needs names supporting exact coefficient arithmetic, equality, leading valuations, and the required support operations. Finite algebra does not by itself supply those oracles. Over rational-function or bounded-denominator Puiseux inputs, finite truncations can be computed using familiar exact symbolic methods; those representations are narrower than the full Hahn field.

The formal lifting proofs are support-controlled in a precise relative sense. After the finite normalizations, record every positive-support parameter used in the inverse and residue-simple equations. The corresponding formal solution supports lie in the monoid generated by those parameter supports. Monomial rescalings add their explicitly known shifts. This is *not* an assertion that the output uses only the supports of the raw moment errors: inverse separations, normalized units, and starting coefficients contribute to the normalized parameters as well.

10.2 What the accompanying program checks

This directory contains `code/verify.py`, which uses exact SymPy arithmetic and writes a human-readable verification log. Its tests cover:

1. the Hermite cardinal conditions and the matrix identity $J^{-1}J = I$ for several finite rational configurations;
2. the exact cofactor correction system, moment annihilator, cross-numerator identity, and recovered finite Laurent coefficients for explicit rational perturbations;
3. the two-node formula (39), its discriminant and boundary collision, and the displayed power-series coefficients;
4. the three-node cancellation comparison, its nonlinear middle-weight obstruction, and the rank-two threshold arithmetic;
5. a directed graph with a negative edge but positive cycles, verifying shortest-path potentials and the path lower bounds.

There is no floating-point evidence in these checks. They detect sign, index, and algebraic mistakes in concrete instances. They do not prove arbitrary-rank summability, the general quantified theorems, or any novelty claim. Those rest on the mathematical arguments, with their stated classical prerequisites.

The recorded run, `data/verification.txt`, passed 10,052 exact assertions under Python 3.13.5 and SymPy 1.14.0; the count includes finite graph-walk enumeration and is not a count of independent results. The program first deletes that file and then rewrites it, recording the Python and SymPy versions on its second line, so it should be run on a copy of this directory. When the report joined the collection, a run on such a copy under Python 3.14.4 and SymPy 1.14.0 reproduced every line of the record except that version line. The program keeps the manuscript's letters (Section 1.4),

and `data/requirements.txt` pins the tested SymPy version.

10.3 A possible Lean development

A formalization can be separated into finite algebra and valuation lemmas. The first layer is the Prony moment functional, the cofactor basis, its orthogonality matrix, and the finite Padé identity. These are field-level statements. The second layer adds a valuation, proves integral-polynomial and divided-difference estimates, and packages residue-simple lifting as an explicit assumption. It then proves the sharp uniform theorem without any reference to a proper class.

The third layer instantiates the lifting assumption for Hahn fields via strong formal evaluation, and transfers finite families to the actual surreal carrier through the normal-form bridge. The Hermite inverse layer can reuse the repository’s finite interpolation machinery: the collection’s formalization ledger `docs/FORMALIZATION.md` maps the Hermite and Lagrange interpolation clauses of `polynomial:thm:crt` to checked Lean declarations over fields. The graph certificate is finite ordered-group combinatorics plus matrix scaling. The exact lattice theorem needs the multivariable residue-simple lifting lemma, not an ordinary normed-space inverse-function theorem.

No Lean proof of this article is included or claimed. The repository’s existing build and axiom audit concern its own imported declarations, not the new statements here. A complete mechanization would need to match each hypothesis, particularly the exact model order, strict inequalities, and universe sizes.

11 Boundaries of the results and further questions

The uniform theorem resolves the precise finite reconstruction question posed here, including optimality at each configuration. It is not presented as a solution to a named previously published open problem. The primary research claim is a proposed new exact refinement with complete proofs, subject to independent verification and a broader priority search.

Three distinctions are particularly important. First, the threshold is optimal for retaining all strict nearest-neighbour labels, not for the bare existence of any atomic representation. Second, the weight bound $\kappa - 2d_i - \delta_i$ is uniformly valid above the sharp threshold, whereas the improved cancellation bound with q_i is proved for the tangent map and, under a separate finite coefficient criterion, for the whole nonlinear uncertainty set. Third, the directed-path theorem is a sufficient nonuniform certificate, not the exact geometry of all allowed error ideals.

Several focused problems remain. An exact nonuniform precision region would have to retain algebraic correlations among the entries of the cofactor correction matrix, rather than just their valuations. A sharp criterion for when the entire inverse image is its Hermite lattice may be weaker than (32). Confluent configurations, where several atoms merge and derivative evaluations replace simple evaluations, require a different local model; they are not covered by declaring a zero separation valuation to be finite. Finally, infinite configurations need a separate summability hypothesis on the moment data and are outside this finite theorem.

These questions are stated as research directions, not asserted to be known open problems in the published literature. The present finite results already supply a useful answer: the information needed to resolve surcomplex atoms is governed by a weighted, pointwise collision profile, and the exact worst moment direction can be written down without solving any nonlinear equation.

A Dependency and novelty ledger

Statement or ingredient	Status and dependencies
Hahn normal forms and finite workspace localization	Standard background. The interpretation in actual surreal numbers uses Conway normal forms, as discussed in [5, 7].
Positive-support evaluation	Classical support principle, deduced here from Higman’s word theorem [6]. It does not assert sequential convergence.
Prony uniqueness, Hankel factorization, Hermite duals	Classical finite algebra; proved here for completeness and credited to the Prony/interpolation background [1, 2].
Theorem 1.1 and Proposition 6.1	Principal proposed refinement: exact arbitrary-rank, pointwise threshold, nonlinear weight control, and a universal highest-moment witness. Depends on cofactor linear algebra, simple lifting, and divided differences.
Proposition 7.1	Exact valuation consequence of the classical Hermite inverse, with explicit cancellation invariant. Not a claim to have invented the inverse matrix.
Proposition 7.2	Support-based arbitrary-rank polynomial adaptation of the established differential precision principle [4].
Corollary 7.3	Exact precision-set application, under its extra smallness criterion. Not asserted to hold throughout the entire sharp-threshold region.
Theorem 8.1	Proposed sufficient nonuniform certificate. It uses cycle potentials and strongly summable matrix inversion. Its conditions are not claimed necessary.
Verification program	Exact checks of finite examples only. Not a formal proof, a proof of infinite support assertions, or a priority certificate.
Section 1.4	Added on placement in the collection: provenance, notation, the meaning of “moment”, and cross-references to neighbouring reports. No mathematical claim.
Remark 6.2	Added on placement: an elementary comparison of Θ with the threshold $T(P)$ of <code>polynomial:thm:stability</code> [8], with two small examples. No new theorem, and no coefficient-side universal sharpness claimed.

B Reproducibility and source audit

The repository was inspected through its connected GitHub interface. The pinned commit is recorded in Section 1.3. At that commit the inspected `docs/README.md` listed twenty-six principal reports; the `docs/surcomplex/polynomial-algebra/README.md` described its sixty-one-page merged report, including its coefficient threshold, Hermite interpolation, and support results. The full main-report collection, the eighteen separate ZIPs then in `docs/new`, and removed historical sources were not exhaustively compared. Indexed no-match results are reported as observations, not as proof that the terms or results occur nowhere in the repository. These are statements about the pinned snapshot. The collection has since grown, this report being one of the additions, and the archives then in `docs/new` have been placed into reports (Section 1.3).

The primary literature comparison used the authors’ arXiv manuscripts [1, 2, 3, 4, 5] and publisher metadata for the journal references. The exact arbitrary-rank theorem stated here was not identified in that material. Keyword searches combining Prony with Hahn, surreal, or valued fields did not yield a matching primary theorem, but several search results were broad or irrelevant; they cannot establish comprehensive absence. The manuscript therefore does not certify priority. The manuscript modified no repository file; joining the collection added only this report’s own directory.

Compile from this directory with

```
latexmk -pdf -halt-on-error -interaction=nonstopmode article.tex
python code/verify.py
```

Run the second command on a copy of the directory: it deletes and rewrites `data/verification.txt` (Section 10). The tested SymPy version is pinned in `data/requirements.txt`. The LaTeX bibliography is internal. No proprietary fonts, external images, or downloaded papers are required or bundled. The verification script uses SymPy; the PDF and its proofs do not require executing that script.

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- [7] V. Reshetnikov and contributors, *Surreal: Lean formalization and research reports*, GitHub repository, inspected 22 September 2026, commit 4cf691c7d951e037739d32d9f5c387dcce724f3c. Pinned repository snapshot. Specifically the documentation catalogue, formalization-ledger excerpts, and polynomial-algebra report README. Research reports are not treated as independently refereed literature.

- [8] *Surcomplex Polynomial Algebra: Root Geometry, Newton Profiles, Multiscale Stability, and Residue Duality*, report `docs/surcomplex/polynomial-algebra` of this collection; in particular `polynomial:thm:stability`, `polynomial:ex:sharpdisc`, `polynomial:cor:discprecision` and `polynomial:thm:crt`. Cited on placement; an AI-assisted research draft, not refereed literature.
- [9] *Surcomplex Spectral Theory*, report `docs/surcomplex/spectral-theory` of this collection; its section on Hankel square-class profiles. Cited on placement; an AI-assisted research draft.
- [10] Report `docs/surreal/hahn-valued-measures-and-probability` of this collection, on Hahn-valued measures, moment representation and probability. Cited on placement; an AI-assisted research draft.
- [11] Report `docs/surcomplex/hahn-herglotz-positivity` of this collection, on Hahn–Herglotz normalization and trigonometric moments. Cited on placement; an AI-assisted research draft.
- [12] *Surcomplex Analysis*, report `docs/surcomplex/analysis` of this collection; in particular `d:thm:moments`. Cited on placement; an AI-assisted research draft.